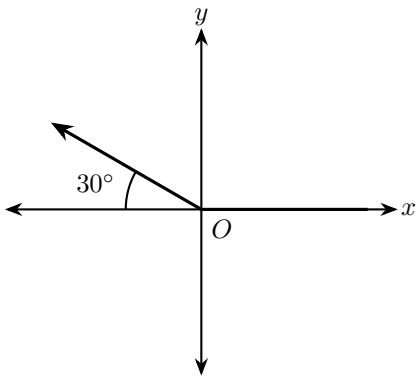
Name _____ Period _____ Date _____

7-1 Lesson Quiz: The Unit Circle

short form

1. Select **all** the possible measures for the angle shown.

- ☐ **A.** $-\frac{4\pi}{3}$
- ☐ **B.** $-210°$
- ☐ **C.** $\frac{5\pi}{6}$
- ☐ **D.** $210°$
- ☐ **E.** $510°$



Show how you checked one of the choices you selected:

_____

_____

3. Find the measure of an angle that is positive and less than $360°$ in standard position for each reference angle.

$68°$ in quadrant II: _____° Work: _____

$75°$ in quadrant IV: _____° Work: _____

$22°$ in quadrant III: _____° Work: _____

4. Convert the angle measures $\frac{\pi}{6}$ radians and $\frac{5\pi}{3}$ radians to degrees.

$\frac{\pi}{6}$ radians = _____° Work: _____

$\frac{5\pi}{3}$ radians = _____° Work: _____

Scoring: Question 3 and Question 4 are 10 points each. Question 1 is a bonus of up to 5 points (all three correct boxes, none wrong). Total out of 20.

7-1 Lesson Quiz, short form: answer key

The numbers are the Savvas item numbers (TE printed page 353A). The TE prints no standards codes; the skill codes are the Realize "Skills Review and Practice" codes from the TE item analysis.

1. **B, C, E.** The terminal side is $30°$ above the negative x-axis, so the angle is $180° - 30° = 150°$. B: $150° - 360° = -210°$. C: $\frac{5\pi}{6} \cdot \frac{180°}{\pi} = 150°$. E: $150° + 360° = 510°$. Distractors: A, $-\frac{4\pi}{3} = -240°$, is coterminal with $120°$, not $150°$; D, $210°$, is in Quadrant III. Savvas item 1, DOK 1, skill F84 (Examples 1 and 3). **Bonus 5:** B, C and E with no extra box; 3 for two correct boxes and none wrong; 0 otherwise.

3. $112°$**;** $285°$**;** $202°$**.** Quadrant II: $180° - 68° = 112°$. Quadrant IV: $360° - 75° = 285°$. Quadrant III: $180° + 22° = 202°$. Savvas item 3, DOK 1, skill F84 (Example 1, Parts A and B). **10 points:** 3 per blank, plus 1 for writing the subtraction or addition for at least one blank (Example 1 writes each angle as $180° \pm$ or $360° -$ the reference angle).

4. $30°$**;** $300°$**.** $\frac{\pi}{6} \cdot \frac{180°}{\pi} = 30°$; $\frac{5\pi}{3} \cdot \frac{180°}{\pi} = \frac{5 \cdot 180°}{3} = 300°$. Savvas item 4, DOK 1, skill F85 (Example 3A). **10 points:** 5 each, of which 2 for the work (multiplying by $\frac{180°}{\pi}$).

Items left off this form (available for a retake or Form B): Savvas item 2, a reference angle of $35°$ with the terminal side in Quadrant IV, possible positive and negative measures (answer A, $325°$ and $-35°$; DOK 1, F84; Example 1); Savvas item 5, a carousel horse on a circle of radius 15 ft through $\frac{2\pi}{3}$ radians (answer 31 ft, since $15 \cdot \frac{2\pi}{3} = 10\pi \approx 31.4$; DOK 2, F84; Example 4, which the fast packet cuts). The TE answer template for item 5 prints "$\frac{2\pi}{3}$ radians = __ ft"; on a Form B write "arc length = __ ft".

IXL replacement: SmartScore 70 or above on 7CV, BRP and EDC replaces this quiz with 20/20.